

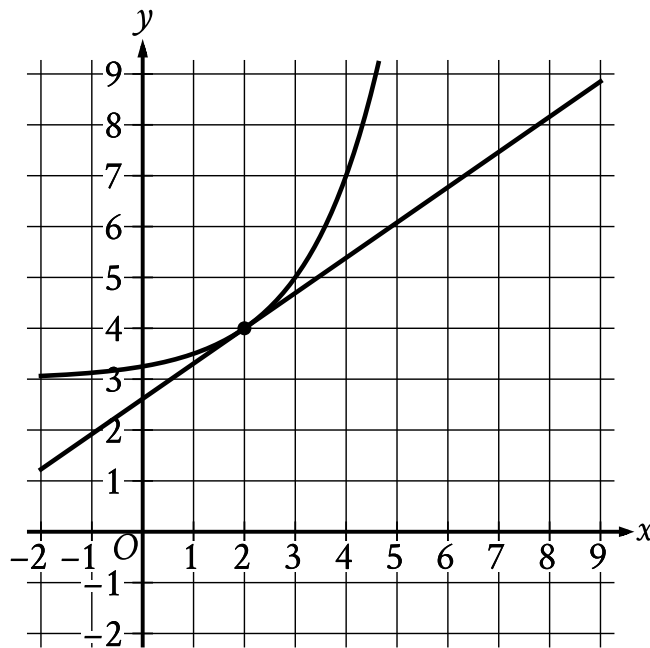
● EASY

● ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

02

10 questions · ~15 min



- For the curve in the system:
 - Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to point (0 comma 3.25).
 - In quadrant 1, the curve trends up sharply.
 - The curve passes through the following points:
 - (1 comma 3.5)
 - (2 comma 4)
 - (4 comma 7)
- For the line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (2 comma 4)
 - (7 comma 7.5)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

- A. 0,0
- B. 0,2
- C. 2,4
- D. 4,0

Q2

GRID-IN

ADVANCED MATH

$$x^2 = 2222$$

What is the positive solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ADVANCED MATH

$$x + y = 12$$

$$y = x^2$$

If (x, y) is a solution to the system of equations above, which of the following is a possible value of x ?

- A. 0
- B. 1
- C. 2
- D. 3

Q4

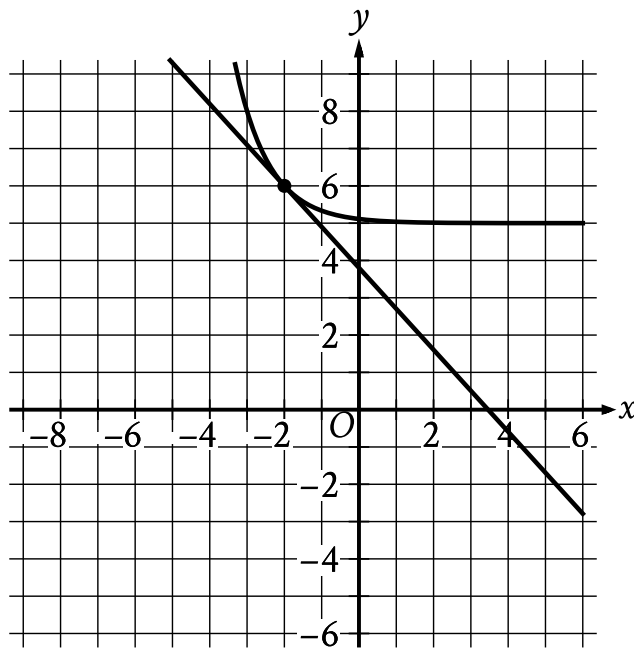
ADVANCED MATH

$$y = 5x + 4$$

$$y = 5x^2 + 4$$

Which ordered pair x, y is a solution to the given system of equations?

- A. 0, 0
- B. 0, 4
- C. 8, 44
- D. 8, 84



- For the line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (negative 2 comma 6)
 - approximately (0 comma 3.8)
- For the curve in the system:
 - Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2:
 - The curve trends down sharply to the point (negative 2 comma 6).
 - The curve then trends down gradually to the y axis.
 - In quadrant 1, the curve trends down gradually.
 - As x increases, the curve approaches the line y equals 5.
 - The curve passes through the following points:
 - (negative 3 comma 8)
 - (negative 2 comma 6)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

- A. 6, 0
- B. -2, 6
- C. 0, -2
- D. 0, 0

Q6

ADVANCED MATH

$$\frac{r}{38} = s + t$$

The given equation relates the quantities r , s , and t . Which equation correctly expresses r in terms of s and t ?

- A. $r = s + t + 38$
- B. $r = s + t - 38$
- C. $r = 38s + t$
- D. $r = \frac{s+t}{38}$

Q7

ADVANCED MATH

$$q - 29r = s$$

The given equation relates the positive numbers q , r , and s . Which equation correctly expresses q in terms of r and s ?

- A. $q = s - 29r$
- B. $q = s + 29r$
- C. $q = 29rs$
- D. $q = -\frac{s}{29r}$

Q8

ADVANCED MATH

$$k^2 - 53 = 91$$

What is the positive solution to the given equation?

- A. 144
- B. 72
- C. 38
- D. 12

$$7m = 2n + p$$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives m in terms of n and p ?

A. $m = \frac{2n + p}{7}$

B. $m = 2n + p$

C. $m = 2n + p - 7$

D. $m = 2 - n - p - 7$

Q10

GRID-IN

ADVANCED MATH

$$x - 5 = 10$$

What is one possible solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.